

a) $f(x,y) = \sqrt{y-x-2}$

$$y-x-2 \geq 0$$

$$x-y+2 \leq 0$$

$$x-y \leq -2$$

6) Hallar las derivadas parciales de primer orden

$$a) f(x, y) = (2x - 3y)^3$$

$$\frac{\partial z}{\partial x} = 3(2x - 3y) \cdot 2 = 6(2x - 3y)^2$$

$$\frac{\partial z}{\partial y} = 3(2x - 3y)^2 \cdot (-3) = -9(2x - 3y)^2$$

$$b) f(x, y) = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\frac{\partial z}{\partial x} = \frac{-\frac{y}{x^2}}{1 + \left(\frac{y}{x}\right)^2} = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial z}{\partial y} = \frac{\frac{1}{x}}{1 + \left(\frac{y}{x}\right)^2} = \frac{x}{x^2 + y^2}$$

$$c) f(x, y) = e^{-x} \sin(x + y)$$

$$\frac{\partial z}{\partial x} = -e^{-x} (\sin(x + y) + e^{-x} \cos(x + y))$$

$$\frac{\partial z}{\partial y} = e^{-x} \cos(x + y)$$

$$d) f(x, y) = e^{xy} \ln y$$

$$\frac{\partial z}{\partial x} = y e^{xy} \ln y$$

$$\frac{\partial z}{\partial y} = x e^{xy} \ln y + \frac{e^{xy}}{y}$$

$$e) f(x, y) = \cos^2(3x - y^2)$$

$$\frac{\partial z}{\partial x} = 2 \cos(3x - y^2) \cdot (-\sin(3x - y^2)) \cdot 3 = -6 \sin(3x - y^2) \cos(3x - y^2)$$

$$\frac{\partial z}{\partial y} = 2 \cos(3x - y^2) \cdot (-\sin(3x - y^2)) \cdot (-2y) = 4y \sin(3x - y^2) \cos(3x - y^2)$$

$$f) f(x, y) = \frac{x+y}{xy-1}$$

$$\frac{\partial z}{\partial x} = \frac{1(xy-1) - [y(x+y)]}{(xy-1)^2} = \frac{\cancel{xy} - 1 - \cancel{xy} - y}{(xy-1)^2} = \frac{-y-1}{(xy-1)^2}$$

$$\frac{\partial z}{\partial y} = \frac{1(xy-1) - [(x+y)(x)]}{(xy-1)^2} = \frac{\cancel{xy} - 1 - x^2 - \cancel{xy}}{(xy-1)^2} = \frac{-x^2-1}{(xy-1)^2}$$

$$g) f(x, y) = \sqrt{x^2 + y^2}$$

$$\frac{\partial z}{\partial x} = \frac{1}{2} (x^2 + y^2)^{-1/2} \cdot 2x = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial z}{\partial y} = \frac{1}{2} (x^2 + y^2)^{-1/2} \cdot 2y = \frac{y}{\sqrt{x^2 + y^2}}$$

7) Hallar las derivadas parciales de segundo orden

$$a) z = x^2 y + \cos y + y \sin x \quad \text{Rta.: } \frac{\partial^2 z}{\partial x^2} = 2y - y$$

$$\frac{\partial z}{\partial x} = 2xy + y \cos x$$

$$\frac{\partial^2 z}{\partial x^2} = 2y - y \sin x$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2x + \cos x$$

$$\frac{\partial z}{\partial y} = x^2 - \sin y + \sin x \quad \frac{\partial^2 z}{\partial y^2} = -\cos y \quad \frac{\partial^2 z}{\partial x \partial y} = 2x + \cos x$$

$$z = tg^{-1}\left(\frac{y}{x}\right)$$

$$\frac{\partial z}{\partial x} = \frac{-\frac{y}{x^2}}{1 + \left(\frac{y}{x}\right)^2} = \frac{-y}{x^2 + y^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{0 - [-y(2x)]}{(x^2 + y^2)^2} = \frac{2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{-1(x^2 + y^2) - [-y(2y)]}{(x^2 + y^2)^2} = \frac{-x^2 - y^2 + 2y^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\frac{\partial z}{\partial y} = \frac{\frac{1}{x}}{1 + \left(\frac{y}{x}\right)^2} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{0 - [x(2y)]}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{x^2 + y^2 - (x(2x))}{(x^2 + y^2)^2} = \frac{x^2 + y^2 - 2x^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$c) w = ye^{x^2-y}$$

$$\frac{\partial w}{\partial x} = 2xye^{x^2-y} \quad \frac{\partial^2 w}{\partial x^2} = 4x^2ye^{x^2-y} \quad \frac{\partial^2 w}{\partial x \partial y} = 2x(e^{x^2-y} + y(-e^{x^2-y})) = 2xe^{x^2-y} - 2xye^{x^2-y}$$

$$\frac{\partial w}{\partial y} = e^{x^2-y} + y(-e^{x^2-y}) = e^{x^2-y} - ye^{x^2-y} \quad \frac{\partial w}{\partial y \partial x} = 2xe^{x^2-y} - 2xye^{x^2-y}$$

8) Sea $f(x, y) = 2x + 3y - 4$. Encontrar la pendiente de la recta tangente a esta superficie en el punto $(2; -1)$ y que se encuentra en: